UNIVERSITY OF JOHANNESBURG

Department of Applied Information Systems | Development Software 2B

STEP 3A | STUDENT DEBUGGING ANSWER SHEET

Graded Lab: 40 marks | 90 minutes | Submit with corrected source code

Student name: ____Athenkosi silinga____ Student number: __224108331___

Use this document to record your OWN diagnosis and evidence. Save as PDF or DOCX. Screenshots may be placed after the relevant answer or submitted as a separate evidence folder.

Part A — Campus Tasks | ChatGPT allowed | 20 marks

Complete A1–A4 (5 marks each). For each task: diagnosis (1), implementation (3), verification (1). Provide a prompt transcript separately.

A1. Task loading

Cause of problem (1 mark):

extractRecords reads response.items, but the bundled JSON stores the array at data.tasks. The property is undefined, Array.isArray fails, and the function returns [], so no tasks display.

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File/function changed and summary of code correction (3 marks):

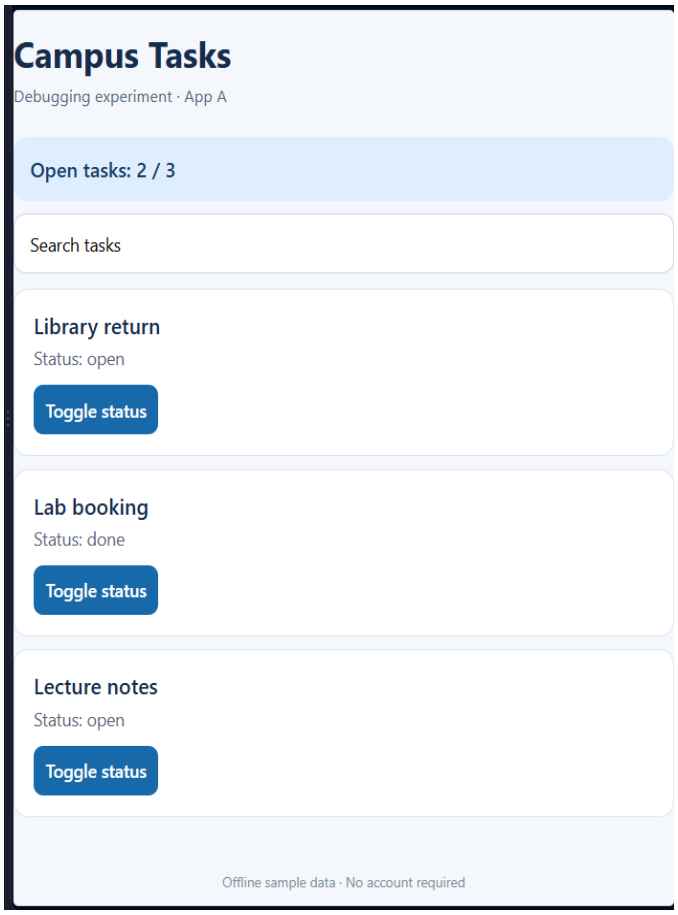
Step1_Campus_Tasks_Student_App/campus-tasks/src/logic.cjs, function extractRecords. Changed the property path from response?.items to response?.data?.tasks, keeping the Array.isArray fallback

Code correction: const values = response?.data?.tasks;

._____

_____

Verification: expected result, actual result, screenshot filename (1 mark):



A2. Case-insensitive search

Cause of problem (1 mark):

The query is converted to lower case but item.title is not, so titles that contain capital letters e.g. "Library return"

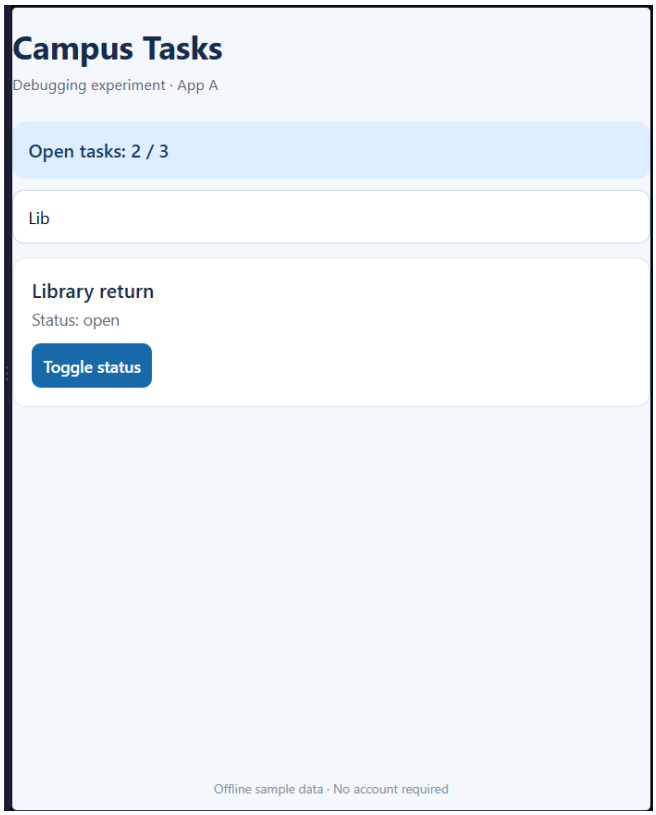
never match, and an upper-case query such as "LIB" finds nothing.

File/function changed and summary of code correction (3 marks):

File: Step1_Campus_Tasks_Student_App/campus-tasks/src/logic.cjs, function filterRecords. Added .toLowerCase() to the title before .includes(q), so both sides are lower case.

Code correction : String(item.title || '').toLowerCase().includes(q)

Verification: expected result, actual result, screenshot filename (1 mark):



A3. Status update

Cause of problem (1 mark):

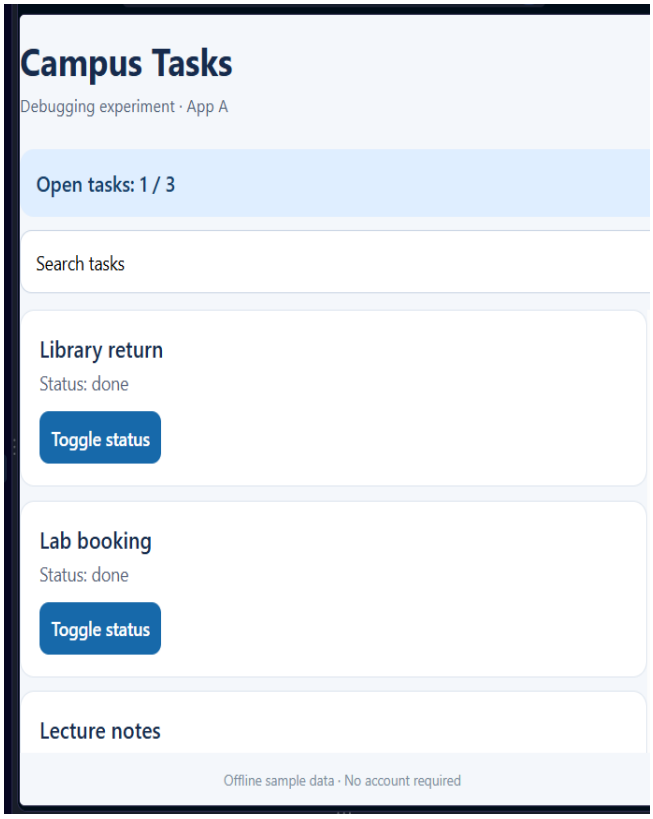
toggleRecord finds the right id but sets status to item.status, i.e. to its own current value, so nothing ever changes.

File/function changed and summary of code correction (3 marks):

Step1_Campus_Tasks_Student_App/campus-tasks/src/logic.cjs, function toggleRecord. For the item whose id matches, return a copy with status flipped between 'open' and 'done'; all other items are returned unchanged.

Code correction : { ...item, status: item.status === 'open' ? 'done' : 'open' }

Verification: expected result, actual result, screenshot filename (1 mark):



A4. Open-task count

Cause of problem (1 mark):

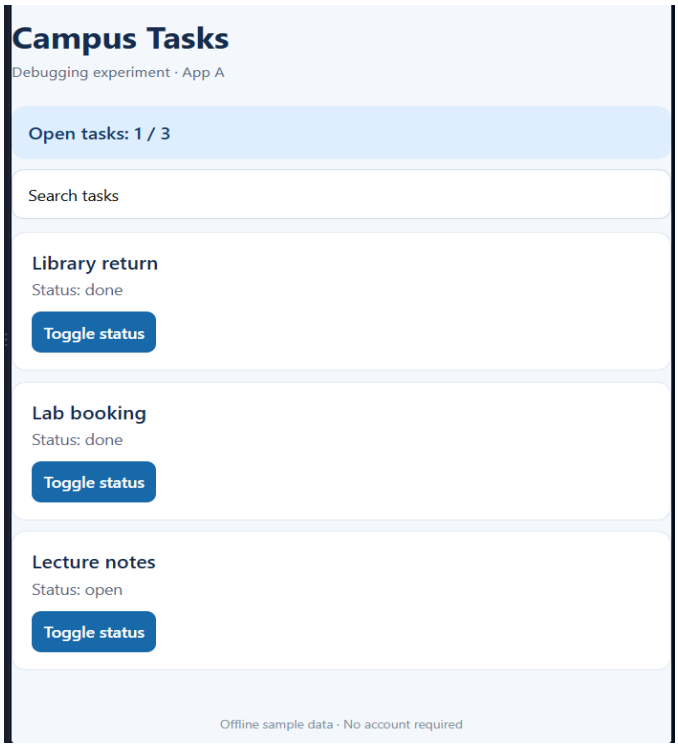
_countOpen filters for status === 'done', so the dashboard counts completed tasks instead of open tasks.

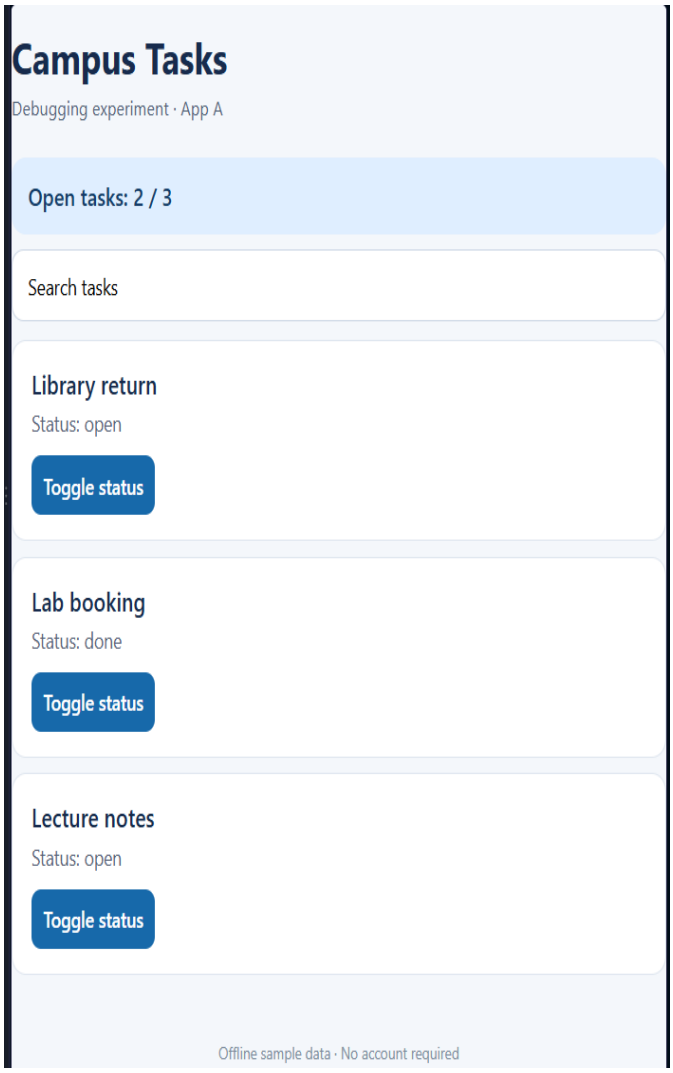
File/function changed and summary of code correction (3 marks):

Step1_Campus_Tasks_Student_App/campus-tasks/src/logic.cjs, function countOpen. Changed the filter condition from 'done' to 'open'

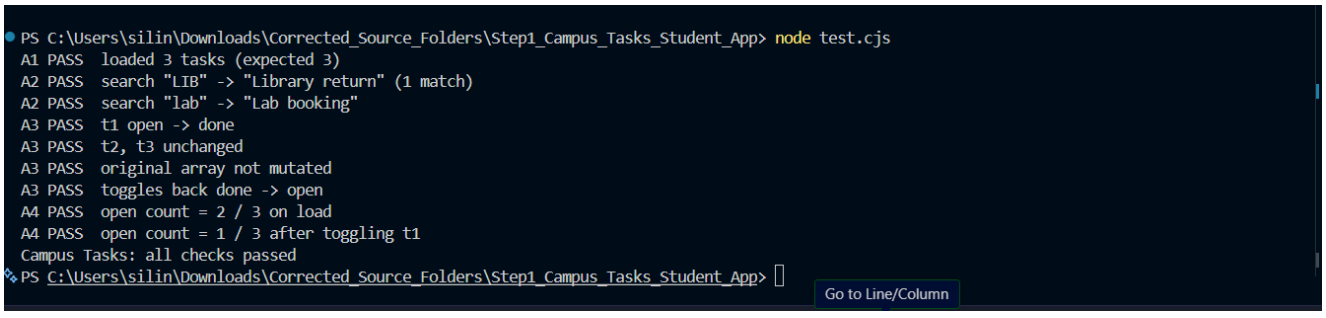
Code correction: records.filter(item => item.status === 'open').length

Verification: expected result, actual result, screenshot filename (1 mark):


Campus Tasks
Debugging experiment · App A
Open tasks: 1 / 3
Search tasks
Library return
Status: done
Toggle status
Lab booking
Status: done
Toggle status
Lecture notes
Status: open
Toggle status
Offline sample data · No account required



NODE A TEST:



Part B — Service Tickets | NO AI | 15 marks

Complete B1–B3 (5 marks each). Do not consult ChatGPT, earlier transcripts, or other AI systems.

B1. Ticket loading

Cause of problem (1 mark):

extractRecords reads response.items, but the JSON nests the array at data.tickets. The result is undefined, so the function returns [] and no tickets display

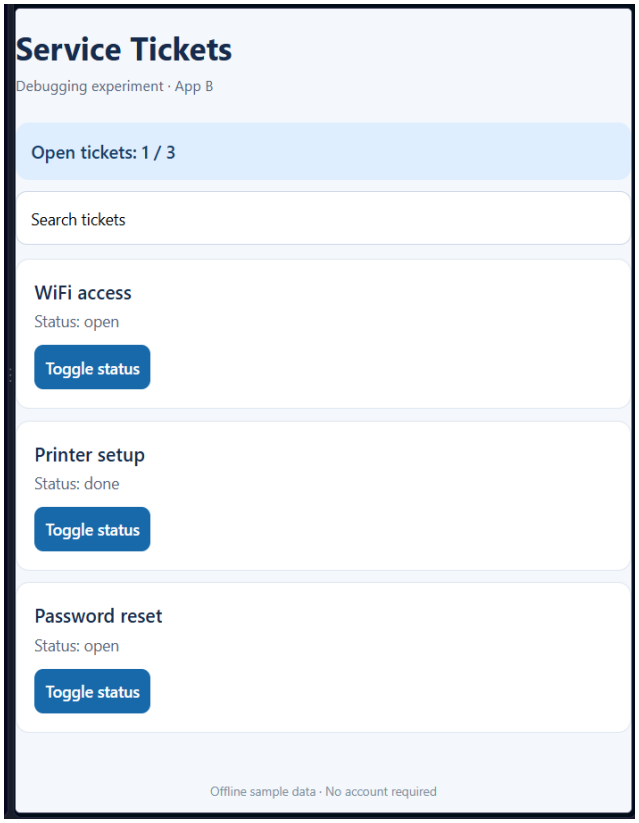
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File/function changed and summary of correction (3 marks):

Step1_Service_Tickets_Student_App/service-tickets/src/logic.cjs, function extractRecords. Changed response?.items to response?.data?.tickets.

Corrected code: const values = response?.data?.tickets;

Verification: expected result, actual result, screenshot filename (1 mark):



B2. Case-insensitive search

Cause of problem (1 mark):

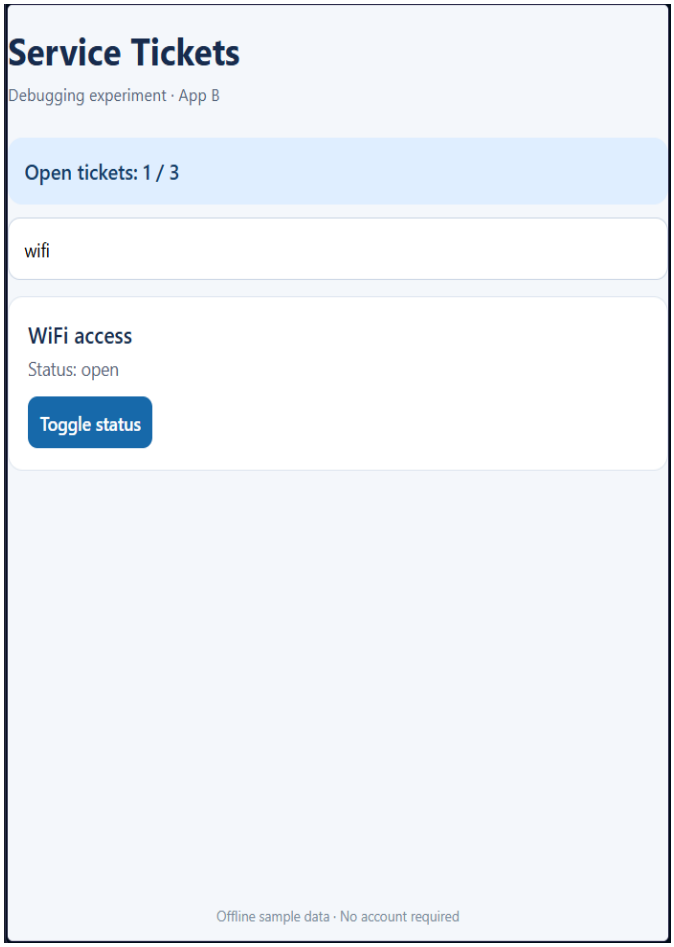
The query is lower-cased but item.subject is not, so "WiFi access" cannot match "wifi" or "WIFI".

File/function changed and summary of correction (3 marks):

Step1_Service_Tickets_Student_App/service-tickets/src/logic.cjs, function filterRecords. Added .toLowerCase() to the subject before .includes(q).

Correct code: String(item.subject || '').toLowerCase().includes(q)

Verification: expected result, actual result, screenshot filename (1 mark):



B3. Status update

Cause of problem (1 mark):

toggleRecord sets status to item.status (itself), so no ticket ever changes. The item.id === id check was already correct, so only the selected ticket should change

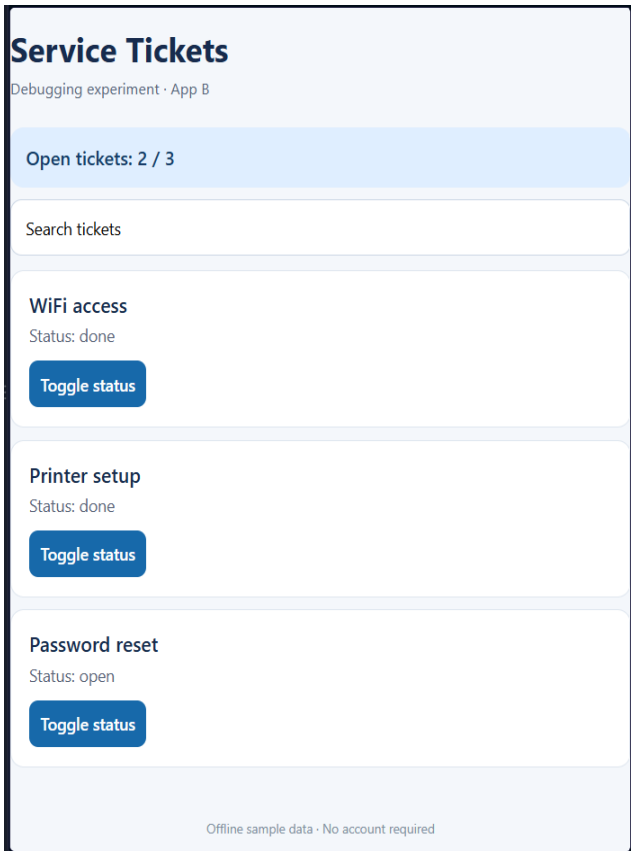
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File/function changed and summary of correction (3 marks):

Step1_Service_Tickets_Student_App/service-tickets/src/logic.cjs, function toggleRecord. For the matching id, return a copy with status flipped between 'open' and 'done'; every other ticket is returned unchanged.

Correct code : { ...item, status: item.status === 'open' ? 'done' : 'open' }

Verification: expected result, actual result, screenshot filename (1 mark):



NODE B TEST:

```
Step1_Service_Tickets_Student_App > service-tickets > JS App.js > ...
   1   import React, { useEffect, useMemo, useState } from 'react';
Problems  Output  Terminal  Ports

    Directory: C:\Users\silin\Downloads\apps\Step1_Ser
    vice_Tickets_Student_App\service-tickets


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        2026/10/26     22:52                src
-a----        2026/10/26     22:52           3082 App.
                                                 js
-a----        2026/10/26     22:52            181 app.
                                                 json
-a----        2026/10/26     22:52            494 pack
                                                 age.
                                                 json
-a----        2026/10/26     22:52            424 READ
                                                 ME.m
                                                 d
-a----        2026/10/26     23:04           1275 test
                                                 .cjs


PS C:\Users\silin\Downloads\apps\Step1_Service_Tickets_Student_App\service-tickets> node test.cjs
B1 PASS  loaded 3 tickets (expected 3)
B2 PASS  search "WIFI" -> "WiFi access" (1 match)
B2 PASS  search "printer" -> "Printer setup"
B3 PASS  s1 open -> done
B3 PASS  s2, s3 unchanged
B3 PASS  original array not mutated
B3 PASS  toggles back done -> open
Open count shown by unfixed countOpen on load: 1 / 3 (correct value is 2 / 3 -> defect for C3)
Service Tickets: all checks passed
PS C:\Users\silin\Downloads\apps\Step1_Service_Tickets_Student_App\service-tickets>
```

Part C — Independent code explanation | NO AI | 5 marks

C1 (2 marks). Explain why you must inspect the response structure (e.g. data.tickets) before reading the array, and what happens if you use the wrong property.

The shape of a response is decided by the API or JSON file, not by my code, so I must inspect it first to find where the array actually is (here data.tickets, not items or the top level). If I read the wrong property I get undefined; Array.isArray then fails and my fallback returns an empty array. The app does not crash or show an error, it just displays an empty list, which makes the bug easy to miss and hard to trace even though the data exists.

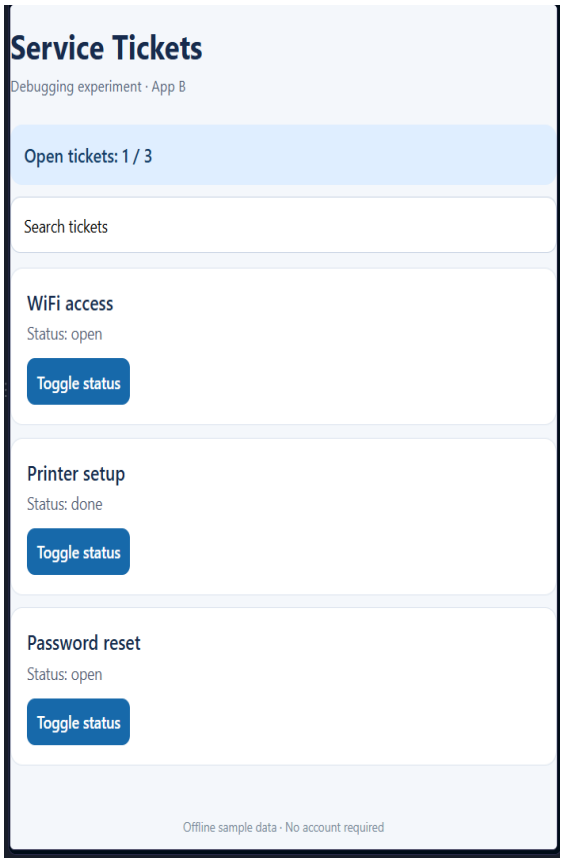
C2 (2 marks). Explain how to toggle only the selected ticket and why React state updates should preserve unchanged records.

I use map over the tickets and compare each ticket's id with the selected id. Only the matching ticket is replaced by a copy with the status flipped ({ ...item, status: newValue }); every other ticket is returned as the same object, so unrelated tickets cannot change.

React detects changes by comparing references. If I mutated the existing object or array, the reference would not change, React could skip the re-render and the screen would show stale data. Creating a new array and a new object only where something changed gives React a clear signal, keeps useMemo dependencies working and avoids side effects on shared state.

C3 (1 marks). Describe a test to check that an open-ticket summary remains correct after a status change (you do not need to fix this defect in Part B).

The sample data has 3 tickets, 2 open and 1 done so the summary should read 2 / 3. After each status change I compare the displayed number with the number of cards showing "Status: open": toggle an open ticket to done and expect 1 / 3, toggle it back and expect 2 / 3, toggle the done ticket and expect 3 / 3. A unit test on countOpen with known arrays (2 open + 1 done must return 2) would confirm the same. In the unfixed app the summary shows 1 / 3 on load and never matches the cards, which proves the defect.



Final submission checklist

☐ Corrected Campus Tasks folder (exclude node_modules)

☐ Corrected Service Tickets folder (exclude node_modules)

☐ This completed answer sheet

☐ Part A ChatGPT transcript, only for Part A

☐ Screenshots or test evidence for each debugging task

I confirm AI was used only for Part A.

Signature: _____ Date: ____26-10-2026_____